

BIOGRAPHICAL SKETCH

Provide the following information for the key personnel and other significant contributors in the order listed on Form Page 2.
Follow this format for each person. **DO NOT EXCEED FOUR PAGES.**

NAME Kumiko Yamaguchi, M.D., Ph.D.		POSITION TITLE Research Scientist	
eRA COMMONS USER NAME			
EDUCATION/TRAINING <i>(Begin with baccalaureate or other initial professional education, such as nursing, and include postdoctoral training.)</i>			
INSTITUTION AND LOCATION	DEGREE <i>(if applicable)</i>	YEAR(s)	FIELD OF STUDY
Kagawa University, Kagawa, Japan	M.D.	1991	Medicine
Kagawa University, Kagawa, Japan	Ph.D.	1999	Physiology

NOTE: The Biographical Sketch may not exceed four pages. Items A and B (together) may not exceed two of the four-page limit. Follow the formats and instructions on the attached sample.

A. Positions and Honors. List in chronological order previous positions, concluding with your present position. List any honors. Include present membership on any Federal Government public advisory committee.

Positions

1991 - 1992 Orthopedist, Department of Orthopedics
Kagawa Medical University, Kagawa, Japan

1992 - 1995 Research Fellow, Molecular Genetics Unit
University of Cambridge, U.K.

1999 - 2000 Research Fellow, Dr. Sydney Brenner Lab
Molecular Sciences Institute, Berkeley, CA

2000 - 2003 Senior Scientist
Lynx Therapeutics, Inc., Hayward, CA

2003 - present Research Scientist
Molecular Sciences Institute, Berkeley, CA

Other Experience and Professional Memberships

None

B. Selected peer-reviewed publications (in chronological order). Do not include publications submitted or in preparation.

1. Compston, J.E., Garrahan, N.J., Croucher, P.I., Wright, C.D., and Yamaguchi, K. Quantitative analysis of trabecular bone structure. *Bone*, 14, 187-192, 1993.
2. Yamaguchi, K., Croucher, P.I., and Compston, J.E. Comparison between the lengths of individual osteoid seams and resorption cavities in human iliac crest cancellous bone. *Bone and Mineral*, 23, 27-33, 1993.
3. Compston, J.E., Yamaguchi, K., Croucher, P.I., Garrahan, N.J., Lindsay, P.C., and Shaw, R.W. The Effects of Gonadotropin-Releasing Hormone Agonists on Iliac Crest Cancellous Bone Structure in Women with Endometriosis. *Bone*, 16(2), 261-267, 1995.

4. Yamaguchi, K., Grant, J., Noble-Jamieson, G., Jamieson, N., Barnes, N.D., and Compston, J.E. Hypercalcaemia in Primary Oxalosis: Role of Macrophage induced bone resorption and effects of treatment with Pamidronate. *Bone*, 16(1), 61-67, 1995
5. Yan, H., Yang, J., Marasco, J., Yamaguchi, K., Brenner, S., Collins, F., and Karbon, W. Cloning and functional expression of cDNAs encoding human and rat pancreatic polypeptide receptors. *Proc. Natl. Acad. Sci. USA*, 93, 4661-4665, 1996
6. Yamaguchi, K., Yamaguchi, F., Miyamoto, O., Sugimoto, K., Konishi, R., Hatase, O., and Tokuda, M. Calbrain, a novel two EF-hand calcium-binding protein that suppresses Ca²⁺/Calmodulin-dependent protein kinase II activity in the brain. *J. Biol. Chem.* 274, 3610-3616, 1999
7. Yamaguchi, K., Yamaguchi, F., Tokuda, M., Konishi, R., Hatase, O., and Brenner, S. Molecular cloning of NPY receptor genes using highly degenerate primers from fugu rubripes. *J. Biochem. Mol. Biol. & Biophys.* 2, 293-299, 1999
8. Yamaguchi, K., Yamaguchi, F., Miyamoto, O., Hatase, O., and Tokuda, M. The reversible change of GluR2 RNA editing in gerbil hippocampus in course of ischemic tolerance. *J. Cereb. Blood Flow Metab.* 19, 370-375, 1999
9. Yamaguchi, F., Yamaguchi, K., Tokuda, M., and Brenner, S. Molecular cloning of EDG-3 and N-Shc genes from the puffer fish, *Fugu rubripes*, and conservation of synteny with the human genome. *FEBS Letter*, 459, 105-110, 1999
10. Yamaguchi, F., Yamaguchi, K., and Tokuda, M. Presenilin-1 protein specifically expressed in Leydig cells with its expression level increased during rat testis development. *Int. J. Biochem Cell Biol.* 32(1), 81-87, 2000
11. Yamaguchi, F., Yamaguchi, K., Tai, Y., Sugimoto, K., and Tokuda, M. Molecular cloning and characterization of a novel phospholemmann-like protein from rat hippocampus. *Molecular Brain Res.*, 86, 189-192, 2001
12. Yamaguchi, F., Yamaguchi, K., and Brenner, S. Genomic organization and linkage analysis of GABA_A α 1 subunit gene in the puffer fish *FUGU rubripes*. *J. Biochem. Mol. Biol. & Biophys.* 5, 172-179, 2001
13. Burbulis, I., Yamaguchi, K., Carlson, R., and Brent, R. Using protein-DNA chimeras to detect and count small numbers of molecules. *Nature Methods* 2005, 2:31-37.

C. Research Support. List selected ongoing or completed (during the last three years) research projects (federal and non-federal support). Begin with the projects that are most relevant to the research proposed in this application. Briefly indicate the overall goals of the projects and your role (e.g. PI, Co-Investigator, Consultant) in the research project. Do not list award amounts or percent effort in projects.

Ongoing Research Support

None

Completed Research Support

None